DATATHON 2026

Problem Statement & Proposed Solution

Theme: Real-Time AI for Critical Decisions

Our objective: Build a real-time AI-powered decision-support system that analyzes transactional events as they happen, identifies suspicious or abnormal behavior, calculates risk, and recommends an immediate action.

1. Problem Statement

Financial systems process a large number of transactions continuously. Traditional analysis may identify fraudulent or abnormal behavior only after a transaction has already been completed or after multiple suspicious events have accumulated. This delay can increase financial loss and make intervention harder.

The problem is therefore to create a system that can analyze transactional data in real time, detect anomalies or suspicious patterns, estimate the level of risk, and support a fast operational decision before a potentially harmful transaction causes greater impact.

2. What We Are Going to Build

We will build a Real-Time AI Fraud & Risk Decision Engine. Every incoming transaction is treated as a live event. The system evaluates the transaction against relevant behavioral and contextual signals, produces a risk score, explains the important risk factors, and recommends one of three actions:

- ALLOW — transaction appears normal.
- REVIEW — transaction requires additional verification.
- BLOCK — transaction has a sufficiently high risk level to stop or escalate.

3. Example Scenario

A customer normally makes transactions around ₹2,000–₹5,000. A new transaction arrives for ₹87,500 at 3:17 AM from a new device and an unusual location. Instead of waiting for a later report, the system evaluates these signals during the transaction flow.

Signal	Observed Value
Transaction amount	₹87,500
Normal customer average	₹2,400
Device	New / previously unseen
Time	03:17 AM
Location	Unusual
AI risk score	91 / 100
Recommended action	BLOCK / MANUAL REVIEW

4. How the System Works

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Transaction Event
↓
Real-Time Event Processing
↓
Feature & Behavior Analysis
↓
AI / Anomaly Detection
↓
Risk Score + Explanation
↓
ALLOW / REVIEW / BLOCK
↓
Live Dashboard + Alert
```

5. What the AI Will Analyze

- Transaction amount compared with the customer's normal behavior.
- Transaction frequency and unusual transaction velocity.
- New or unusual device information.
- Location deviation or unusual geography.
- Unusual transaction time.
- Recent failed or suspicious attempts.
- Sudden changes in customer behavior.
- Combinations of multiple weak signals that together indicate higher risk.

6. Main Features

Real-Time Transaction Stream — Receive and process transactions as live events.

Risk Score — Convert transaction signals into a 0–100 risk score.

Anomaly Detection — Identify behavior that differs from an expected pattern.

Decision Engine — Turn the risk result into ALLOW, REVIEW or BLOCK.

AI Explanation — Show understandable reasons behind the risk result.

Live Alerts — Immediately highlight high-risk events.

Transaction Details — Display the complete context and decision factors.

Risk Analytics — Show transaction volume, risk distribution and decision trends.

Transaction History — Store processed events and decisions for later analysis.

7. Live Dashboard

The main interface will behave like a real-time command center. New transactions appear automatically, their risk scores update, and critical events are highlighted without manually refreshing the page.

Dashboard Area	Purpose
Live Transactions	Latest incoming transactions and current decisions
Risk Score	Current risk level from 0–100
Alerts	Critical/high-risk transactions requiring attention
Decision Stats	ALLOW / REVIEW / BLOCK counts
Risk Analytics	Charts showing patterns and trends
Transaction Details	Signals, explanation and recommended action

8. Technology Architecture

```
Transaction / IBM Z Integration
↓
Node.js + Express Event Layer
↓
Risk & Anomaly Engine
↓
AI Decision / Explanation Layer
↓
MongoDB ←→ Socket.io
↓
React + Tailwind Live Dashboard
```

Technology	Role
React	Real-time frontend dashboard
Tailwind CSS	Dashboard UI and responsive styling
Node.js + Express	Backend and transaction processing

Socket.io	Real-time event delivery to the dashboard
MongoDB	Transaction, customer and risk history
AI / ML	Anomaly detection, scoring and decision support
IBM Z / z/OS	Enterprise transactional integration / demonstration

9. Real-Time Data Without n8n

n8n is not required for the core system. A backend event/simulation service can generate clearly labelled demo transactions and send them through the same real-time pipeline. This allows the complete system to be demonstrated consistently. Where an actual transactional source is available, it can replace or complement the simulator.

Important: simulated events must be clearly identified as DEMO/SIMULATION DATA and must not be presented as real-world fraud incidents.

10. Why This Is a Decision-Support System

The goal is not simply to show a fraud dashboard. The important part is the decision loop: the system receives an event, evaluates it immediately, identifies risk, explains the relevant signals, and recommends an action while the transaction is still time-sensitive.

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EVENT → INTELLIGENCE → RISK → EXPLANATION → DECISION → ACTION
```

11. Final Demo

The final demonstration will show normal transactions being allowed, followed by intentionally generated high-risk demo transactions. The audience will see the event arrive, the risk score appear, the suspicious factors become visible, and the decision change to REVIEW or BLOCK in real time.

Final one-line pitch: “We built a real-time AI decision engine that analyzes transactions as they happen, detects abnormal behavior, explains the risk, and recommends an immediate action before a potential loss occurs.”

12. Scope Boundary

- The system is a prototype / decision-support solution, not a guaranteed fraud detector.
- Risk scores are model outputs and should be treated as decision support.
- Demo-generated transactions are simulations and will be labelled accordingly.
- IBM Z / z/OS integration should be demonstrated meaningfully rather than represented only by branding.
- The primary goal is a reliable end-to-end real-time decision pipeline.

REAL-TIME DATA • AI • RISK • DECISION